Corporate Finance — Mock Exam

Questions Only

> **Instructions:** This mock exam follows the same format and difficulty as the illustrated exam. It covers material from CH3 (Bonds), CH4 (Stocks), CH5–6 (Investment Decisions), CH7–9 (Risk, Return, WACC), CH10 (Capita Selecta), and the ACTUA guest cases. A TI-84 Plus calculator is permitted.

QUESTION TYPE 1 — Multiple Choice

Circle the letter of the single best answer for each question.

Q1. In September 2023, the Belgian government launched a 1-year state bond (the "Van Peteghem bond") at a historically high subscription rate. Suppose the government had instead issued a **3-year** bond with a nominal value of **€1,000**, an annual coupon rate of **3.3%** (paid annually), and redemption at par. If the required yield-to-maturity (YTM) for such a bond is **1.5%** per year, what is the fair price of this bond today?

a. €1,032.45

b. €1,042.38

c. **€1,052.42**

d. €1,062.51

Q2. Consider a 2-year corporate bond with a nominal value of €100, an annual coupon rate of **6%** (paid annually), redeemed at par. The current YTM is **4%** per year. What is the **Macaulay duration** of this bond?

a. 1.87 years

b. 1.91 years

c. **1.94 years**

d. 1.98 years

Q3. The Belgian authentication startup **NextGrowth NV** is expected to report earnings per share (EPS) of **€6.00** next year. The company applies a **50% payout ratio** and has a return on equity (ROE) of **18%**. The cost of equity is **12%**. What is the **Present Value of Growth Opportunities (PVGO)** of NextGrowth NV?

a. €25.00

b. €35.00

c. **€50.00**

d. €65.00

Q4. A portfolio consists of two assets A and B. Asset A has an expected return of **10%** and a standard deviation of **15%** and represents **30%** of the portfolio. Asset B has an expected return of **16%** and a standard deviation of **22%** and represents **70%** of the portfolio. The correlation between A and B is **$\rho = 0.3$**. Which of the following **best quantifies the risk** of this portfolio?

a. Standard deviation = 12.95%

b. Standard deviation = 20.15%

c. **Variance = 2.990%**

d. Variance = 4.990%

Q5. Company **HEX NV** has **4 million shares** outstanding at a market price of **€10 per share**, and **€40 million** in market-value debt. Its cost of debt is **6%**, the tax rate is **25%**, and the covariance of HEX's returns with market returns is **0.02**. The risk-free rate is **2%**, the standard deviation of market returns is **10%**, and the expected market return is **7%**. What is HEX's **WACC**?

a. 7.25%

b. **8.25%**

c. 9.25%

d. 10.25%

QUESTION TYPE 2 — Short Closed Questions

Answer each question briefly and to the point.

Q2.1 In the course we discussed the Belgian pension reform plan (2017), which allows employees to 'buy off' years of higher education for a government pension supplement at a known upfront cost and a known yearly benefit upon retirement.

Explain conceptually (without calculation) which **uncertain factors** determine whether one should participate in this pension scheme. For each factor, explain the **direction** of its effect on the attractiveness of the scheme.

Q2.2 In September 2023, the Van Peteghem bond raised €21.9 billion. However, a second attempt in February 2024 failed, partly because of an **inverted yield curve**.

(a) Explain what a **yield curve** is and what a **normal vs. inverted** yield curve signals about the economy.

(b) Explain why an **inverted yield curve** made it less attractive for the Belgian government to issue another 1-year bond in 2024.

Q2.3 The course introduced two competing capital structure theories: the **trade-off theory** and the **pecking order theory**. Explain the core insight of each theory.

Q2.4 Explain what a **real option** is in the context of capital budgeting. Also explain (a) why the presence of real options **increases** the NPV of an investment project compared to a standard DCF analysis, and (b) why real options become **more valuable** as uncertainty increases.

Q2.5 Explain what **beta (β)** is, how it is measured in practice, and why it is used specifically — rather than total standard deviation — to estimate a company's cost of equity via the CAPM.

QUESTION TYPE 3 — Large Exercise

You are the CFO of **BioTech NV**, a listed pharmaceutical company, and are evaluating a capital investment to upgrade the company's production line. You have collected the following financial data:

Financial information on BioTech NV:

Item	Value
Number of outstanding shares	8 million
Book value per share	€5
Market value per share	€9
Market value of debt	€72 million
Cost of debt	6%
Tax rate	25%
Covariance BioTech share and market return	0.03
Expected market return	10%
Standard deviation of market return	15%
Risk-free rate (annual ST government bond)	4%

Two investment proposals have been received:

- **Proposal X:** Capital investment (CAPEX) of **€600,000** in new production equipment with an economic life of **5 years**. Annual turnover will increase by **€240,000** and annual operating costs will increase by **€60,000**.
- **Proposal Y:** Capital investment (CAPEX) of **€800,000** in upgraded production infrastructure with an economic life of **8 years**. Annual turnover will increase by **€260,000** and annual operating costs will increase by **€55,000**. This proposal also requires an additional **€10,000/year** in maintenance costs.

In both cases, BioTech NV will depreciate the investment **straight-line** over the economic life. BioTech NV is a profitable company; profits are taxed at **25%**.

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a. Estimate BioTech NV's **cost of capital (WACC)**. Explain step by step what you are doing and what your calculated figures mean. Also explain the **role of WACC** in an investment decision.

b. Which proposal do you prefer? Make an informed decision using the **appropriate evaluation method**, given that BioTech NV is a going-concern company. Explain why you use this method, show all calculations, and interpret your results. Make additional assumptions if necessary.

c. Suppose that as a result of the chosen proposal, BioTech's production becomes significantly more efficient and inventory levels are expected to **decrease by €50,000** at the start of the project. Do you expect this to have an impact on the financial value generated by the project? Why yes or no? Explain without calculations.

d. Suppose that instead of straight-line depreciation, BioTech uses **declining balance (degressive) depreciation** for the chosen machinery investment. Do you expect this to affect the financial value generated? Why yes or no? Explain without calculations.

QUESTION TYPE 4 — ACTUA Cases & Capita Selecta

Q4.1 — Gu & Lev (2011): "Overpriced Shares, Ill-Advised Acquisitions, and Goodwill Impairment"

In the course, we discussed the academic paper by Gu & Lev (2011), which establishes an empirical link between share overpricing, acquisition behaviour, and subsequent goodwill write-offs.

(a) Briefly explain the **main thesis** of Gu & Lev (2011). What is the **root cause** of goodwill write-offs according to the authors? Use the following terms in your answer: share overpricing, acquisition with overpriced shares, goodwill, goodwill write-off.

(b) Explain how the findings of Gu & Lev (2011) relate to the concept of **NPV** and the concept of **PVGO** covered in the course. In particular, explain why acquiring with overpriced shares can represent a **negative-NPV investment** and what role managerial psychology plays.

Q4.2 — Belgian Pension Reform (ACTUA1): Time Value of Money in Practice

In the course, we analysed the Belgian government pension reform plan (2017). Under the scheme, an employee can pay **€2,025 today** (maximum, after tax benefits) to receive **€750 extra per year** upon retirement for the rest of their life.

(a) Explain in your own words why **time value of money (TVM)** is a critical factor when evaluating this pension scheme. Make reference to the opportunity cost of capital.

(b) For a person aged **35**, retiring at **65** (t = 30 years) and expecting to live **20 years** after retirement, the course material shows that the break-even annual return is approximately **5.18%**. If this person can consistently earn **7% per year** on alternative investments, should they participate in the pension scheme? Explain your reasoning qualitatively — no calculation required.

End of exam.